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MATH 014 - Intro. to Data Science
28 April 2028

Crime Around the Clock: A Seasonal Analysis of DC Crime Incidents During 2025

Objective

The purpose of this project is to understand overall crime occurrences within DC. I am trying to identify the temporal and spatial patterns in crime across Washington, D.C. to understand when and where crime takes place. Additionally, I am trying to explore how crime clusters around specific times, places, or seasonal trends. Overall, my goal with this dataset is to understand DC crime and the patterns it follows.

Introduction

On August 11, 2025, the National Guard was deployed to DC as a way to combat the immense crime the city was apparently facing. Returning to DC for school during this time was quite a jarring experience. There was a heated debate concerning the safety and reported crime rates within DC; city officials argued that reported crime had fallen sharply while others, including Donald Trump, argued that DC was in the middle of a federal emergency related to crime. This dataset, Crime Incidents in 2025 by Open Data DC, is analytically compelling because of its relevance to the matter and it contains direct reports of crime within DC with time, dates, wards, and other valuable entries. This dataset is a complete record of all the crime reports that occurred in DC during 2025. This dataset has a shape of 24,152 rows and 25 columns. For each entry, the crime's location, time, method, offense, and other supporting details are described.

Method

A. Data Understanding

Before manipulating the data, I ensured that I had a basic understanding of its attributes. After opening the notebook, I imported all the necessary libraries and packages I would need for analysis later on. I started my data understanding my reading/uploading the dataset. Then, I examined the first and last five rows of the dataset. Moving on, I discovered the shape of the dataset and learned the names of all the columns. After that, I used the `df.info()` function to review the data types of each column as well as the `df.select_dtypes()` function. This is when I realized that the DISTRICT and WARD column were numerical data types instead of categorical; later on, in the cleaning section, I explain why I converted them. Lastly, I did `df.describe()` to see the summary statistics of the dataset. All of the information from that was unusable because the numerical data were actually used for geographical markers. For instance, it does not make sense to have the mean of latitude. Also, I searched for missing values and duplicates within the dataset. There were missing values within the END_DATE, BID, OCTO_RECORD_ID, DISTRICT, and PSA columns and there were no duplicates.

B. Data Cleaning

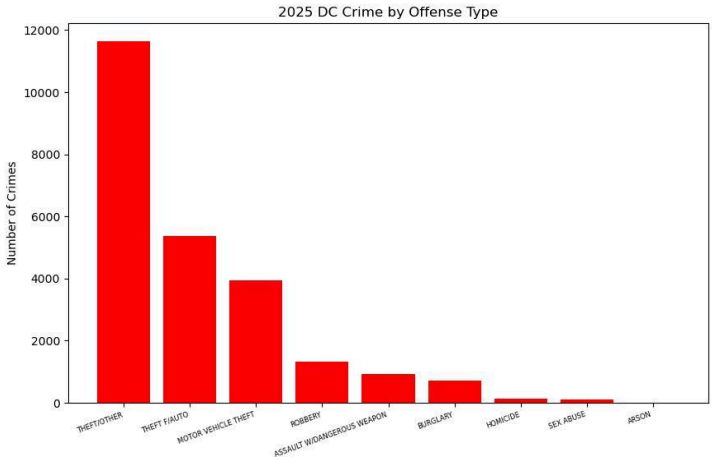
To begin the data cleaning process, I made a copy of the dataset to prevent anything happening to the original. Then, I dropped the

'BLOCK','ANC','PSA','CENSUS_TRACT','NEIGHBORHOOD_CLUSTER','X','Y','CCN','BLOCK_GROUP','BID','XBLOCK','YBLOCK','OBJECTID','OCTO_RECORD_ID' columns because they were either redundant or unnecessary for my analysis. For example, using the ward system over the block names or neighborhood clusters was more efficient because the wards are more geographically encompassing and widely used. Further, the X and Y coordinates were not needed for my visualizations since the latitude and longitude is already given for each entry. Moving forward, I converted the DISTRICT and WARD columns to object data types because in reality the numbers serve a categorical purpose. Since the entries only contained the ward number, Python reads the numbers as numerical data that can be used for calculations. So, to prevent any mishaps, I converted them from number to object data types. After checking the dataset, I then saved the cleaned dataset.

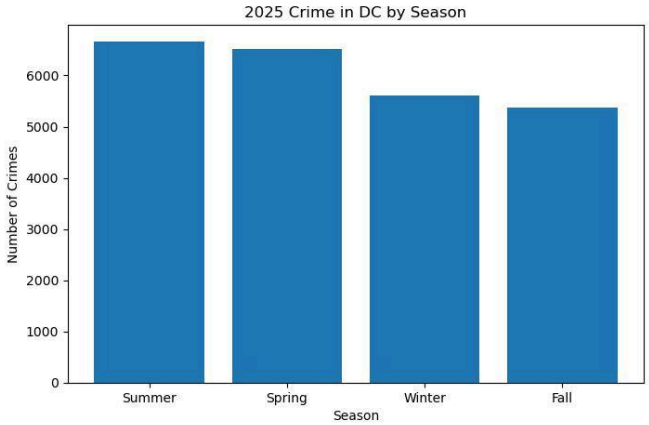
C. Data Preparation and Exploration

Firstly, I conducted a frequency and value count of crime offenses and found that Theft/Other accounts for 48% of total crime in DC. Also, I compared the amount of crime by ward, voting precinct and the top crime in each ward for group-level comparisons. In order to do a temporal and seasonality analysis, I converted the start date column into separate day, month, and hour columns. I chose the start date for this instead of the report date because some witnesses may not report a crime until days after it has been committed. So, using the START_DATE column for this seemed more reliable and accurate. Also, I made a seasons column by defining a function called `get_seasons` to separate each month into its respective season. Having the SEASONS column helped to plot the difference of crimes within each season.

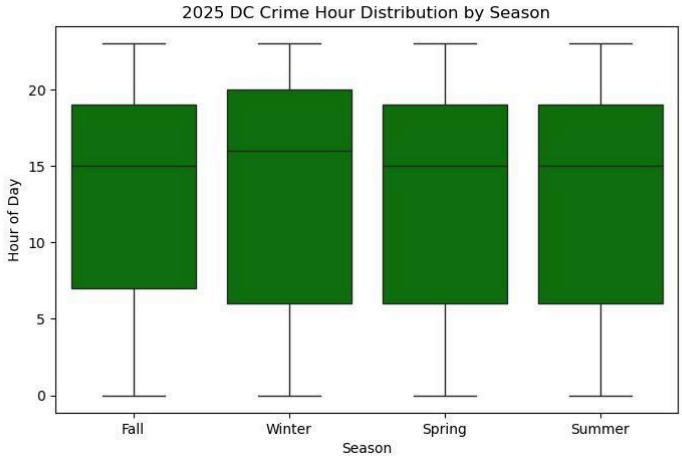
Storytelling: Data Visualization and Interpretation



This first visualization shows the amount of crime as well as the most frequent crime occurring in DC during 2025. This bar chart demonstrates how prevalent overall theft, including THEFT/OTHER, THEFT/AUTO, and MOTOR VEHICLE THEFT, is in DC. The offense that occurred the most in DC during 2025 was Theft/Other. The second-most was Theft/Auto and the third-most was Motor Vehicle Theft. The crime that occurred the least was Arson.

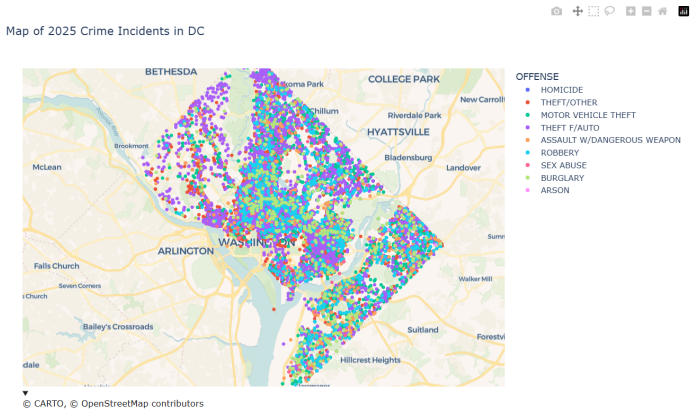


This visualization shows the frequency of crime during each season. I wanted to explore which season had the most crimes occurring, which could then be implied as "the most dangerous season" in DC. Each season had more than 5,000 crime incidents. The season with the most crime is Summer and Spring is almost as equally crime-ridden. The season with the least crime is Fall. From looking at this bar chart, warmer months tend to have more crime activity as opposed to cooler months.

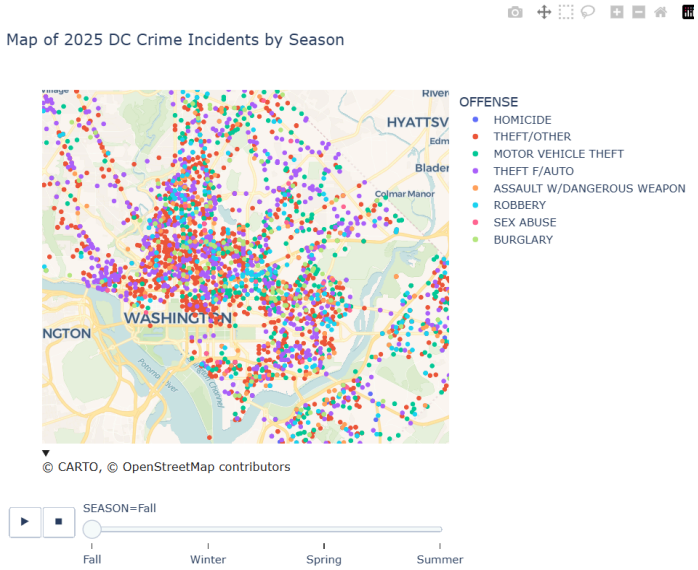


With this visualization, I furthered my analysis of temporal trends to see if there was a specific hourly range that crime happens across seasons to address when crime is the most frequent during the day. Crime timing is fairly consistent throughout the year. The median crime hour for all seasons is around 3:00 PM, so crimes tend to occur most frequently during the afternoon. On the other hand, Winter has the slightly highest median around 4:00 PM. Considering this, crime incidents may happen a bit later in the day during colder months. Crime occurs across nearly the full 24-hour period in every season and there are no strong outliers visible. The distributions for Fall, Spring, and Summer are almost identical, so seasonal weather changes may not strongly affect the hour when crimes occur.

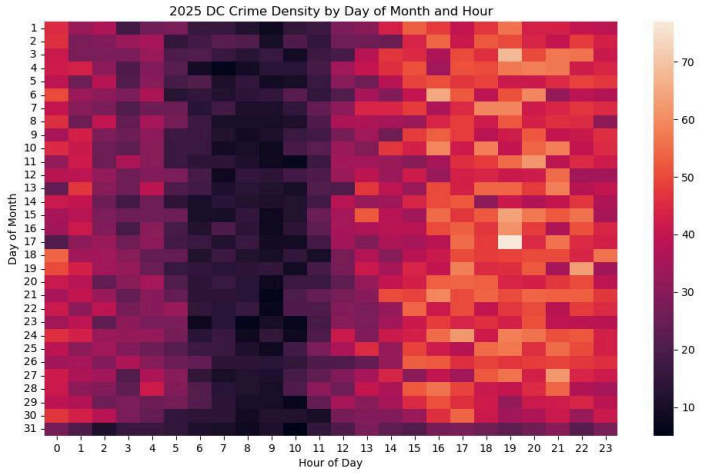
I created this visualization to examine which variable, either time or geographical location, impacted crime volume the most. This visualization allows me to see which ward experiences the most crime as well as when they experience that crime. Also, building off of the season box plot, this visualization helps me to see if the crime levels of the wards are impacted by seasonal crime patterns. For reference, darker colors represent lower crime counts while lighter colors represent higher crime counts. Ward 2 has the highest crime counts overall and Ward 2 experiences consistently high crime activity throughout the year. Ward 5 also shows high crime levels across all seasons, particularly in Spring and Winter. Ward 3 has the lowest crime counts in every season. Summer and Spring tend to show elevated crime levels in many wards and Winter generally shows slightly lower crime counts for most wards. Overall, geographic location seems to have a stronger influence on crime volume than seasonality.



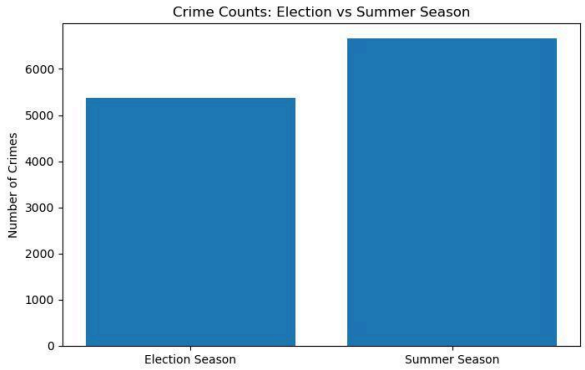
This map visualizes which offense occurs the most as well as where that crime occurs. Using prior knowledge of DC Wards and the DC quadrant system, users can see where different types of crime are concentrated. Crime incidents appear highly concentrated in central DC and in clusters around the NE quadrant and the SE quadrant. The northwest quadrant has the least visible amount of crime, which makes sense as it is known to be an affluent and highly-surveilled area. The map also shows that crime is widespread across wards.



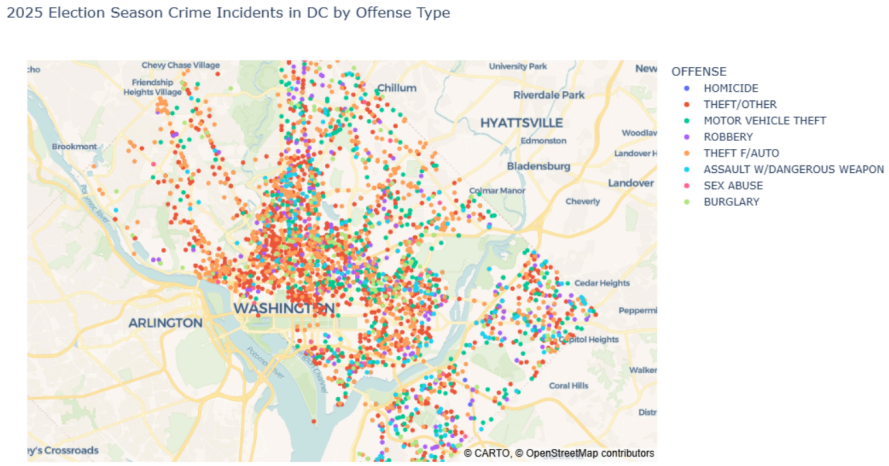
This spatial visualization explores how concentrations of crime and their different offense types change as the seasons progress. Throughout the seasons, Theft/Other and Theft/Auto remains the most prevalent crime across the city. During the Winter, the overall crime is less concentrated and seems more sporadic.



This visualization identifies the peak crime hours across the month and examines when crime is most and least concentrated throughout a typical month. Crime density is lowest during early morning hours, 5 AM–10 AM. Crime density increases during the afternoon and evening hours, 3 PM–10 PM. Peak crime periods seem to be around late afternoon and evening. There is no strong concentration tied to specific days of the month. Time of day is seems to be a stronger predictor of crime density than the specific day of the month.



For my additional analysis, I explore the different crime volumes and patterns during the election season period. For further research, this election season dataset could be used to demonstrate how crime during this time may impact voter disenfranchisement and the casting of ballots. This visualization shows that while the crime count during election season is still high, it does not surpass the Summer season's crime occurrences, which is the season with the highest amount of crime. It can be inferred that election season in DC is relatively safe.



This visualization was created to see the spatial distribution of crime during election season. Visualizing the crime occurrences in certain voting precincts may explain reasons for low voter turnout, voter disenfranchisement, etc. Theft/Other continues to be the most prevalent crime and it is concentrated in central DC. The second-most prevalent crime is Theft/Auto, which is spread out but still concentrated in central DC. For further analysis, I would use a GIS tool to map which offenses occur the most in different voting precincts and how that could affect voter turnout.

Conclusion

The visualizations showed that crime patterns are influenced more by geography and time of day rather than by seasonal change. Crime timing remained relatively stable across seasons with most incidents occurring during afternoon and evening hours. Geographic analysis revealed consistent crime concentration in specific wards like Wards 2 and 5 whereas Ward 3 maintained lower crime levels throughout the year. Crime clusters are concentrated in densely populated and high-activity areas like the central DC. Considering this, it is suggested that location plays a stronger role than

seasonality when it comes to crime frequency. Additionally, crime peaks later in the day rather than during early morning hours and seasonal comparisons showed only moderate variation across the year. However, this analysis has limitations like the reliance on a single year of data and the absence of socioeconomic variables that may influence crime trends. Future research could strengthen the analysis by incorporating multiple years of crime data and demographic indicators to better understand long-term crime behavior in the DC area.

Sources:

"Crime Incidents in 2025." Dc.gov, 2025, opendata.dc.gov/datasets/DCGIS::crime-incidents-in-2025/explore?location=38.904150%2C-77.012050%2C12.

Konstantin Toropin. "DC National Guard Deployment in the Nation's Capital Ordered by Trump Is Extended to Feb. 28." AP News, 6 Nov. 2025, apnews.com/article/national-guard-trump-deployment-washington-cbae3a840cec29b0d361413fbe162e14.

Acknowledgements:

Special thank you to Dr. Nerolu for teaching me the Python language; I know feel more confident about data science after taking her class!